General Aptitude

GATE CSE 2027 · complete formula, rule, method & trap sheet

73 reference boxes · **8** chapters ·

Verbal Aptitude | Numerical Computation | Data Interpretation | Mensuration, Geometry & Probability | Analytical Aptitude | Spatial Aptitude | Problem → Standard Method Playbook | 🎯 Representative GATE-Style Worked Problems | Last-minute one-liners

1 Verbal Aptitude

Grammar essentials GATE tests repeatedly

REFERENCE TABLE

Area	What to watch for
Subject-verb agreement	Singular subject → singular verb, even when other nouns sit in between ("The *list* of items **is** long", not "are")
Tenses	Consistency across a sentence; signal words like "since", "for", "yet" hint at the correct tense
Prepositions	Fixed idiomatic pairings: "married **to**", "different **from**", "capable **of**"
Articles (a/an/the)	"a" before consonant SOUNDS (a university), "an" before vowel SOUNDS (an hour)

WORKED EXAMPLE

Worked example

Q: Choose the correct sentence.
(a) "Neither of the students *have* submitted their assignment."
(b) "Neither of the students *has* submitted their assignment."
A: (b) is correct. "Neither" is a singular indefinite pronoun — it always takes a singular verb ("has"), regardless of the plural noun "students" that follows "of".

Vocabulary: idioms & words-in-context

NOTES

GATE rarely tests rote word-lists; it tests whether you can pick the word that fits the **context** and tone of a sentence.

WORKED EXAMPLE

Worked example

Q: "Despite the heavy rain, the team's spirit remained _____." Choose the best fit: (a) damp (b) undaunted (c) soaked (d) reluctant
A: (b) undaunted — it means "not discouraged", fitting the contrast set up by "Despite". (a) and (c) are literal plays on "rain" (a common distractor trick), and (d) doesn't logically follow from the sentence's structure.

Reading comprehension strategy

KEY POINT

1. Read the QUESTIONS first (skim), so you know what to look for.
2. Skim the passage once for overall structure/argument, don't over-read details yet.
3. For "which of the following is true/false" questions, go back and verify each option against the specific line in the passage — don't rely on memory or general impression.
4. Watch for **absolute words** in options ("always", "never", "all") — passages rarely support such strong claims, making these options likely wrong.

Narrative sequencing

NOTES

You're given 4-5 jumbled sentences and must arrange them into a coherent paragraph.

WORKED EXAMPLE

Worked example

Q: Arrange into a coherent paragraph:
P: As a result, crop yields in the region fell sharply.
Q: The monsoon arrived nearly a month late this year.
R: Farmers had already delayed sowing in anticipation.
S: This further compounded the losses they faced.
A: Q-R-P-S. Start with the triggering event (Q: late monsoon) → the farmers' reaction (R: delayed sowing) → the consequence (P: fell yields) → the compounding effect (S: further losses). Look for pronouns ("this", "as a result") and cause-effect chains — they're the strongest clues to correct order.

Sentence completion — a systematic method

NOTES

Single- and double-blank sentence completion is one of the most repeated 1-mark verbal patterns. Do not try the options one by one — read the **logical signal** first.

REFERENCE TABLE

Signal type	Typical words	What the blank must do
Contrast	but, although, though, despite, however, yet, while, unlike	Be the **opposite** of the clue
Continuation	and, moreover, furthermore, hence, therefore, so, as a result	Carry the **same direction** as the clue
Cause and effect	because, since, thus, consequently	Be the logical cause or result of the other clause
Definition / example	colon, dash, "that is", "for instance"	Restate the clue in other words

KEY POINT

Method: (1) predict a word BEFORE reading the options; (2) in a two-blank sentence fix the blank you are surer of and eliminate options with it; (3) check the polarity (positive or negative) of both blanks; (4) read the finished sentence back once — it must sound natural, not merely grammatical.

WORKED EXAMPLE

Worked example — contrast signal

Q: "Although the plan looked _____ on paper, it proved perfectly workable in practice." (a) practicable (b) sensible (c) unrealistic (d) reliable
A: (c) unrealistic. "Although … perfectly workable" sets up a contrast, so the blank must be the opposite of "workable". The other three all mean roughly "workable", which would make "although" pointless.

WORKED EXAMPLE

Worked example — two blanks

Q: "The witness's account was so _____ that even the lawyer who called her found it hard to _____ it." (a) bizarre … defend (b) clear … accept (c) reliable … trust (d) simple … follow
A: (a). "So … that even her own lawyer found it hard to …" needs a strongly negative first word and a verb of support: "bizarre … defend". (b) and (c) make the difficulty illogical (a clear or reliable account is easy to accept or trust), and (d) "so simple … hard to follow" contradicts itself.

Critical reasoning — arguments, assumptions, strengthen / weaken

NOTES

An **argument** is premises (evidence) plus a conclusion, joined by an **unstated assumption**. Find the conclusion first (clue words: therefore, hence, thus, so, it follows that), then ask: *what must be true for these premises to lead there?*

REFERENCE TABLE

Question type	What to look for	The correct option…
Assumption	The hidden bridge from premise to conclusion	Is **necessary**: negate it and the argument collapses
Strengthen	Support for the bridge	Rules out an alternative cause or confirms the link
Weaken	A gap in the bridge	Offers an **alternative cause**, or shows the evidence is unrepresentative
Inference	What the passage forces	Must be true using ONLY the passage — not merely plausible

KEY POINT

Assumption example: "Profit rose after the firm adopted a four-day week; therefore the four-day week raised profit." The hidden assumption is that *nothing else that affects profit changed at the same time*. Negate it — "something else did change" — and the conclusion no longer follows, which is exactly the test for a necessary assumption.

WORKED EXAMPLE

Worked example — weaken

Q: "After the city made bus travel free for students, attendance at its schools rose 8%. Hence free bus travel raises attendance." Which statement most weakens the argument?
(a) Free travel cost the city a large sum. (b) In the same year every school began serving free midday meals. (c) Students said they liked the free passes. (d) The attendance figures come from the schools' own registers.
A: (b). It supplies an *alternative cause* for the rise, so the bus scheme can no longer be assumed responsible. (a) is about cost, not effect; (c) mildly supports the argument; (d) is irrelevant to causation.

Verbal deduction — what MUST follow

NOTES

Deduction questions give a short passage and ask which statement **must be true**. Use only the passage; ignore real-world knowledge.

GATE FAVOURITE

GATE trap

"All A are B" does **not** give "All B are A" (the converse). Its valid twin is the **contrapositive**: "if not B then not A". Also watch quantity words — "some" never becomes "all", and "most" never becomes "all".

WORKED EXAMPLE

Worked example

Q: "Everyone who completed the safety course received a badge. Ravi has no badge." Which must be true?
(a) Ravi is not an employee. (b) Ravi did not complete the safety course. (c) Everyone with a badge completed the course. (d) Ravi will never receive a badge.
A: (b). The rule is "course → badge"; its contrapositive is "no badge → no course". (a) is not forced — the passage never mentions employment. (c) is the converse. (d) predicts the future, which the passage never does.

Word groups & odd-one-out

NOTES

Find the property shared by three words and use the fourth as the tie-breaker. Test several lenses: **category**, **function**, **degree or intensity**, **part–whole**, **origin**.

REFERENCE TABLE

Group	Odd one out	Reason
Brass, bronze, steel, copper	Copper	Copper is an element; the other three are alloys
Sprint, dash, dart, stroll	Stroll	The first three are fast movement; a stroll is slow (degree)
Lion, tiger, leopard, wolf	Wolf	Big cats versus a canine (biological family)
Square, rectangle, rhombus, circle	Circle	The others are quadrilaterals

2 Numerical Computation

Ratios & Percentages

WORKED EXAMPLE

Worked numerical

Q: A solution has acid and water in ratio 3:7. If 10 litres of water is added, the new ratio becomes 3:10. Find the original quantity of acid.
A:

```
Let acid = 3x, water = 7x (original ratio 3:7)
After adding 10 L water: acid=3x, water=7x+10, new ratio = 3:10

3x / (7x+10) = 3/10
30x = 3(7x+10) = 21x + 30
9x = 30  →  x = 30/9 = 10/3

Acid = 3x = 3×(10/3) = 10 litres
```

FORMULA / RULE

Worked numerical

Q: A shopkeeper marks an item 40% above cost price, then offers a 25% discount on the marked price. Find the net profit/loss percentage.
A:

```
Let cost price = 100.
Marked price = 100 × 1.40 = 140
Selling price (after 25% discount) = 140 × 0.75 = 105

Net result = 105 - 100 = 5  → 5% PROFIT
```

Shortcut formula: Net % change = 40 - 25 - (40×25/100) = 40-25-10 = **+5%** (successive percentage change formula: a + b + ab/100, taking discount as negative).

Powers, Exponents & Logarithms

WORKED EXAMPLE

Worked numerical

Q: If $\log_{10}2 = 0.301$, find $\log_{10}80$.
A:

```
80 = 8 × 10 = 2³ × 10
log(80) = log(2³) + log(10) = 3×log(2) + 1 = 3×0.301 + 1 = 0.903 + 1 = 1.903
```

Permutations & Combinations

KEY POINT

```
Permutation (ORDER matters):   nPr = n! / (n-r)!
Combination (ORDER doesn't matter): nCr = n! / (r!(n-r)!)
```

WORKED EXAMPLE

Worked numerical

Q: In how many ways can a committee of 3 be chosen from 5 men and 4 women, such that at least 1 woman is included?
A:

```
Total ways to choose any 3 from 9 people = 9C3 = 84
Ways with ZERO women (all 3 men from the 5 men) = 5C3 = 10

At least 1 woman = Total - (zero women) = 84 - 10 = 74
```

"At least 1" is almost always easiest via the complement trick: Total minus the one excluded case.

WORKED EXAMPLE

Worked numerical

Q: How many distinct arrangements of the letters in "BANANA" are possible?
A:

```
BANANA has 6 letters: B(1), A(3), N(2)

Arrangements with repeated letters = 6! / (3!×2!×1!) = 720 / 12 = 60
```

Whenever letters repeat, divide the total factorial arrangements by the factorial of each repeated letter's count.

Series & Number Patterns

WORKED EXAMPLE

Worked numerical

Q: Find the next term: 2, 6, 12, 20, 30, ___
A:

```
Differences: 6-2=4, 12-6=6, 20-12=8, 30-20=10  → differences increase by 2 each time
Next difference = 12  →  next term = 30+12 = 42

(Pattern check: each term = n×(n+1): 1×2=2, 2×3=6, 3×4=12, 4×5=20, 5×6=30, 6×7=42 ✓)
```

GATE FAVOURITE

GATE strategy

For number series, always compute first differences, then second differences if the first aren't constant. If differences form a clean arithmetic/geometric pattern, you've found the rule. Also always sanity-check with a closed-form formula (like n×(n+1) above) when the pattern is recognizable — it's faster and safer than extrapolating blindly.

Number system — divisibility, remainders, HCF & LCM

REFERENCE TABLE

Divisor	Test
2, 5, 10	Last digit even / 0 or 5 / 0
3, 9	Digit sum divisible by 3 / by 9
4, 8	Last 2 / last 3 digits divisible by 4 / 8
6	Divisible by both 2 and 3
7	Drop the last digit d and subtract 2d from the rest; repeat (203 → 20 − 6 = 14 ✔)
11	(sum of digits in odd places) − (sum in even places) is 0 or a multiple of 11

KEY POINT

```
Remainder rules    (a + b) mod m = (a mod m + b mod m) mod m
                   (a × b) mod m = (a mod m × b mod m) mod m
Last-digit cycles  2, 3, 7, 8 repeat every 4 powers;  4, 9 every 2;  0, 1, 5, 6 never change
Fermat             a^(p−1) ≡ 1 (mod p)  for prime p not dividing a
HCF × LCM = product of the two numbers          (TWO numbers only)

Greatest number dividing a, b, c and leaving remainders r1, r2, r3 :  HCF(a−r1, b−r2, c−r3)
Least number leaving the SAME remainder r on division by a, b, c  :  LCM(a, b, c) + r
Least number that is k SHORT of a multiple of each of a, b, c      :  LCM(a, b, c) − k
```

WORKED EXAMPLE

Worked numerical — divisibility

Q: For which digit x is the number 2x74 divisible by 9?
A: Digit sum = 2 + x + 7 + 4 = 13 + x must be a multiple of 9 → x = 5 (2574 = 9 × 286).

WORKED EXAMPLE

Worked numerical — last digit

Q: Last digit of 7^{2026}?
A: Powers of 7 end in 7, 9, 3, 1, then repeat (period 4). 2026 = 4 × 506 + 2, so it matches the 2nd entry → **9**.

WORKED EXAMPLE

Worked numerical — remainder

Q: Remainder when 2^{100} is divided by 7?
A: $2^3 = 8 \equiv 1 \pmod 7$. 100 = 3 × 33 + 1, so $2^{100} = (2^3)^{33} \times 2 \equiv 1 \times 2 =$ **2**.

WORKED EXAMPLE

Worked numerical — HCF and LCM

Q: Find the HCF and LCM of 84 and 126 and verify the product rule.
A: 84 = $2^2 \cdot 3 \cdot 7$ and 126 = $2 \cdot 3^2 \cdot 7$. HCF = $2 \cdot 3 \cdot 7$ = 42; LCM = $2^2 \cdot 3^2 \cdot 7$ = 252. Check: 42 × 252 = 10584 = 84 × 126 ✔

Mixtures & alligation

KEY POINT

```
Alligation (mixing a cheaper item of value c with a dearer one of value d to get a mean m):

      cheaper : dearer  =  (d − m) : (m − c)

Replacement: a vessel holds V litres of pure liquid; x litres are drawn off and replaced
by water, n times in all  →  pure liquid left = V × (1 − x/V)^n
```

WORKED EXAMPLE

Worked numerical — alligation

Q: Wine at ₹60 per litre is mixed with wine at ₹80 per litre to give a mixture worth ₹68 per litre. In what ratio?
A: cheaper : dearer = (80 − 68) : (68 − 60) = 12 : 8 = **3 : 2**.

WORKED EXAMPLE

Worked numerical — replacement

Q: A vessel holds 40 L of milk. 4 L is removed and replaced with water; this is repeated twice more (three times in all). How much milk remains?
A: 40 × $(1 - 4/40)^3$ = 40 × 0.9^3 = 40 × 0.729 = **29.16 L** (so 10.84 L is water).

Linear equations & inequalities

KEY POINT

```
Two equations  a1·x + b1·y = c1  and  a2·x + b2·y = c2:
   a1/a2 ≠ b1/b2            → exactly one solution
   a1/a2 = b1/b2 ≠ c1/c2    → no solution (parallel lines)
   a1/a2 = b1/b2 = c1/c2    → infinitely many solutions

Inequalities: multiplying or dividing by a NEGATIVE number reverses the sign.
   |x − a| < b   ⟺   a − b < x < a + b
   |x − a| > b   ⟺   x < a − b   or   x > a + b
```

WORKED EXAMPLE

Worked numerical — simultaneous equations

Q: Solve 3x + 2y = 16 and x − y = 2.
A: From the second, x = y + 2. Then 3(y + 2) + 2y = 16 → 5y = 10 → y = 2, x = **4**. Check: 12 + 4 = 16 ✔

WORKED EXAMPLE

Worked numerical — modulus inequality

Q: How many integers x satisfy $|2x - 3| \le 7$?
A: $-7 \le 2x - 3 \le 7$ → $-4 \le 2x \le 10$ → $-2 \le x \le 5$. The integers −2, −1, 0, …, 5 number **8**.

Progressions (AP, GP) and means

KEY POINT

```
AP:  a_n = a + (n−1)d        S_n = (n/2)·[2a + (n−1)d] = (n/2)·(first + last)
GP:  a_n = a·r^(n−1)         S_n = a(r^n − 1)/(r − 1)
S_∞ = a/(1 − r)   when |r| < 1
For positive numbers: AM ≥ GM ≥ HM; for two numbers  GM² = AM × HM
```

WORKED EXAMPLE

Worked numerical — AP

Q: Find the sum of all multiples of 7 from 1 to 200.
A: The terms are 7, 14, …, 196 — that is 196/7 = 28 terms. S = (28/2)(7 + 196) = 14 × 203 = **2842**.

WORKED EXAMPLE

Worked numerical — GP

Q: Three numbers are in GP with sum 14 and product 64. Find them.
A: Write them as a/r, a, ar. Product = $a^3 = 64 \rightarrow a = 4$. Sum: $4/r + 4 + 4r = 14 \rightarrow 2r^2 - 5r + 2 = 0 \rightarrow r = 2$ or ½. The numbers are **2, 4, 8** (or 8, 4, 2).

Clocks & calendars

KEY POINT

```
Minute hand 6° per minute, hour hand 0.5° per minute → relative speed 5.5° per minute.
Angle between the hands at H:M  =  |30H − 5.5M|     (use 360° − angle if it exceeds 180°)
In 12 hours the hands coincide 11 times, are opposite 11 times and are at right angles 22 times.

Calendar: ordinary year = 52 weeks + 1 odd day;  leap year = 2 odd days.
Leap year: divisible by 4, but a century year only if divisible by 400.
Odd days in 100, 200, 300, 400 years = 5, 3, 1, 0.
```

WORKED EXAMPLE

Worked numerical — clock angle

Q: Angle between the hands at 3:40?
A: |30 × 3 − 5.5 × 40| = |90 − 220| = **130°**.

WORKED EXAMPLE

Worked numerical — calendar

Q: 1 January 2027 is a Friday. Which day is 15 August 2027?
A: Days from 1 Jan to 1 Aug = 31+28+31+30+31+30+31 = 212; add 14 more to reach 15 Aug → 226 days. 226 = 32 × 7 + 2, so move 2 days past Friday → **Sunday**.

Time, speed & distance — trains, relative speed, boats

KEY POINT

```
speed = distance / time        km/h → m/s : × 5/18
m/s → km/h : × 18/5
Equal distances at speeds a and b:   average speed = 2ab/(a + b)   (NOT the simple mean)
Same time, speeds a and b:           average speed = (a + b)/2
For a fixed distance, time ∝ 1/speed — a 25 % speed increase cuts the time by 20 %
```

REFERENCE TABLE

Situation	Distance covered ÷ speed
Train (length L) passes a **pole / man**	L ÷ train speed
Train passes a **platform / bridge** (length P)	(L + P) ÷ train speed
Two trains, **same direction**	(L_1 + L_2) ÷ **difference** of speeds
Two trains, **opposite directions**	(L_1 + L_2) ÷ **sum** of speeds

KEY POINT

Boats and streams. Boat speed in still water = b, stream = s. **Downstream = b + s**, **upstream = b − s**. From the two speeds: **b = (d + u)/2**, **s = (d − u)/2**. Average speed over a round trip is 2du/(d + u), less than b.

WORKED EXAMPLE

Worked numerical

Q: A 100 m train and a 150 m train run in opposite directions at 36 km/h and 54 km/h. How long do they take to cross each other completely?
A: Relative speed = 36 + 54 = 90 km/h = 90 × 5/18 = 25 m/s. Distance = 100 + 150 = 250 m. Time = 250/25 = **10 s**.

Time & work — rates, pipes and wages

KEY POINT

```
Work rate = 1/(days to finish alone);  rates ADD for people working together
A in a days, B in b days together:     time = ab/(a + b)
Pick total work = LCM of the days (units) → each person's units/day becomes an integer
Pipes:  inlet = +rate, outlet/leak = −rate        Wages are shared in proportion to WORK DONE
M₁D₁H₁/W₁ = M₂D₂H₂/W₂   (men × days × hours per day ÷ work is constant)
```

WORKED EXAMPLE

Worked numerical

Q: A fills a tank in 10 h and B empties it in 15 h. With both open, how long to fill the empty tank?
A: Take the tank = 30 units: A adds 3 units/h, B removes 2 units/h → net 1 unit/h → **30 h**.

Profit & loss, averages and ages

KEY POINT

```
Profit % = (SP − CP)/CP × 100  (always on COST unless told otherwise)     SP = CP × (1 ± p/100)
Marked price MP, discount d%:   SP = MP × (1 − d/100)
Successive changes multiply: (1 + a)(1 + b)
Average = sum / count.   Replacing a member changes the SUM by (new − old); so  new sum = old sum − old + new
Weighted average = Σ(nᵢ·xᵢ) / Σnᵢ        Multiplying every value by k and adding c: new average = k·avg + c
Ages: let the younger age be x; "after t years" adds t to EVERY age — the DIFFERENCE of ages never changes
```

WORKED EXAMPLE

Worked numerical

Q: The average of 10 numbers is 15. Each is doubled and then increased by 3. New average?
A: 2 × 15 + 3 = **33** (no need to know the individual numbers).

Sets & Venn diagrams

KEY POINT

```
|A ∪ B| = |A| + |B| − |A ∩ B|
|A ∪ B ∪ C| = |A| + |B| + |C| − |A∩B| − |B∩C| − |A∩C| + |A∩B∩C|
Fill a Venn diagram from the INSIDE out: start with the triple overlap, then "exactly two" regions
(pairwise overlap − triple), then "only one" regions.
Outside all circles = total − union.
```

WORKED EXAMPLE

Worked numerical

Q: In a group of 70, 45 like tea, 40 like coffee and everyone likes at least one. How many like both?
A: 70 = 45 + 40 − both → both = **15**.

Numerical estimation

KEY POINT

Round to friendly numbers, keep the **order of magnitude**, and compare the answer choices — they are usually far apart. Handy benchmarks: 1/8 = 12.5 %, 1/6 ≈ 16.7 %, 1/3 ≈ 33.3 %, 3/8 = 37.5 %, √2 ≈ 1.414, √3 ≈ 1.732, π ≈ 3.14. Check sums with the **last digit** or the **digit-sum (mod 9)** before computing in full.

```
49.8 × 20.1 / 9.95  ≈  50 × 20 / 10 = 100     (exact: 100.6)
Unit digit of 7^45: the last digits of 7ⁿ cycle 7, 9, 3, 1
(length 4); 45 mod 4 = 1 → unit digit 7
```

3 Data Interpretation

Pie charts — the 360°/100% conversion

KEY POINT

```
Degrees in a pie slice = (category's % of total) × 360° / 100
Percentage from degrees = (slice's degrees / 360°) × 100
```

WORKED EXAMPLE

Worked numerical

Q: A pie chart shows a company's expenses: Salaries 40%, Rent 25%, Marketing 15%, Utilities 10%, Other 10%. Total expenses = ₹2,00,000. (a) Find the angle for "Rent". (b) Find the actual amount spent on "Marketing".

A:

```
(a) Angle for Rent = 25% × 360° = 0.25 × 360 = 90°
(b) Marketing amount = 15% × 2,00,000 = 0.15 × 200000 = ₹30,000
```

Bar graphs & tables — classic multi-step question

WORKED EXAMPLE

Worked numerical

Q: A bar graph shows a company's profit (in ₹ lakh) over 4 years: 2021: 50, 2022: 60, 2023: 45, 2024: 75. Find: (a) the year-on-year % change from 2022 to 2023, (b) the average annual profit over all 4 years.

A:

```
(a) % change (2022→2023) = (45-60)/60 × 100 = -15/60 × 100 = -25%  (a 25% DECREASE)

(b) Average = (50+60+45+75)/4 = 230/4 = ₹57.5 lakh
```

Common DI traps

GATE FAVOURITE

GATE angle

- "% change" ALWAYS divides by the ORIGINAL (older) value, never the newer one.
- When comparing two pie charts with DIFFERENT totals, the same PERCENTAGE can represent very different ACTUAL amounts — always convert to absolute values before comparing across charts.
- Read axis labels and units carefully (lakh vs crore, thousands vs millions) — a classic easy-mark-losing mistake under time pressure.

Line graphs, multi-series data & caselets

KEY POINT

Read the scale and the units first (₹ crore? thousands? percent?). Then:

```
% change = (new − old)/old × 100                  (base = the EARLIER year)
Percentage POINTS (difference of two percentages) ≠ percent change:   20 % → 25 % is +5 points but +25 %
Profit % = (revenue − expenditure)/expenditure × 100  (on cost, unless the question says "on revenue")
Ratio questions: compare fractions, don't subtract.
Highest absolute value ≠ highest percentage.
```

WORKED EXAMPLE

Worked numerical — two series

Q: Revenue (R) and expenditure (E) in ₹ crore: 2019: 45 / 30 · 2020: 50 / 42 · 2021: 45 / 40 · 2022: 60 / 48 · 2023: 75 / 55. In which year is the profit % (on expenditure) highest?

A:

```
2019: 15/30 = 50 %   2020: 8/42 ≈ 19.0 %   2021: 5/40 = 12.5 %   2022: 12/48 = 25 %   2023: 20/55 ≈ 36.4 %
Highest percentage → 2019, even though 2023 has the highest absolute profit (₹20 crore).
```

COMMON TRAP

Trap

If A is 20 % more than B, then B is only 16.67 % less than A — the base changes. Two equal successive percentage steps from 200 to 242 are 10 % each (200 × 1.1² = 242), not 21 %.

4 Mensuration, Geometry & Probability

Mensuration — area & perimeter (2D)

REFERENCE TABLE

Shape	Area	Perimeter
Rectangle	l × b	2(l+b)
Triangle	½ × base × height	sum of 3 sides
Circle	πr^2	$2\pi r$ (circumference)
Trapezium	½ × (sum of parallel sides) × height	sum of 4 sides

Mensuration — volume & surface area (3D)

REFERENCE TABLE

Solid	Volume	Total Surface Area
Cube (side a)	a^3	$6a^2$
Cuboid (l,b,h)	l×b×h	2(lb+bh+hl)
Cylinder (r,h)	$\pi r^2 h$	$2\pi r(r+h)$
Cone (r,h,slant l)	$\frac{1}{3}\pi r^2 h$	$\pi r(r+l)$
Sphere (r)	$\frac{4}{3}\pi r^3$	$4\pi r^2$

WORKED EXAMPLE

Worked numerical

Q: A cylindrical water tank has radius 7 m and height 10 m. Find its volume (use π=22/7).

A:

```
Volume = πr²h = (22/7) × 7² × 10 = (22/7) × 49 × 10 = 22 × 7 × 10 = 1540 m³
```

Geometry basics

MUST REMEMBER

Pythagoras theorem: in a right triangle, `(hypotenuse)² = (base)² + (height)²`.
Similar triangles: corresponding angles equal, corresponding sides in the same ratio.

WORKED EXAMPLE

Worked numerical

Q: A ladder 13 m long leans against a wall, its foot 5 m from the wall's base. How high up the wall does the ladder reach?

A:

```
height² = 13² - 5² = 169 - 25 = 144  →  height = √144 = 12 m
```

Probability

KEY POINT

```
P(event) = (number of favorable outcomes) / (total possible outcomes)
P(A or B) = P(A) + P(B) - P(A and B)      [for any two events]
P(A and B) = P(A) × P(B)                   [ONLY if A and B are independent]
```

WORKED EXAMPLE

Worked numerical

Q: Two dice are rolled. Find the probability that the sum is either 7 or 11.

A:

```
Total outcomes = 6×6 = 36
Sum=7: (1,6)(2,5)(3,4)(4,3)(5,2)(6,1) → 6 outcomes
Sum=11: (5,6)(6,5) → 2 outcomes
These are mutually exclusive (can't both happen) → P(7 or 11) = (6+2)/36 = 8/36 = 2/9
```

Elementary Statistics

KEY POINT

```
Mean = (sum of all values) / (number of values)
Median = the middle value when data is sorted (average of 2 middle values if even count)
Mode = the most frequently occurring value
```

WORKED EXAMPLE

Worked numerical

Q: Find the mean, median, and mode of: 4, 8, 6, 5, 8, 9, 8.

A:

```
Mean = (4+8+6+5+8+9+8)/7 = 48/7 ≈ 6.86

Sorted: 4,5,6,8,8,8,9  (7 values, middle = 4th value)
Median = 8

Mode = 8 (appears 3 times, more than any other value)
```

5 Analytical Aptitude

Deduction vs Induction

KEY POINT

Deduction
Reasoning from general rules to a specific, GUARANTEED conclusion. "All birds have feathers. A sparrow is a bird. ∴ A sparrow has feathers." — if the premises are true, the conclusion MUST be true.

KEY POINT

Induction
Reasoning from specific observations to a general (probable, not guaranteed) pattern. "Every sparrow I've seen has feathers. ∴ probably all sparrows have feathers." — likely, but not logically certain.

Syllogisms — the classic GATE pattern

WORKED EXAMPLE

Worked numerical

Q: Statements: "All engineers are logical. Some logical people are creative." Conclusion: "Some engineers are creative." Is this conclusion valid?
A: NOT valid. "All engineers are logical" only tells you engineers are a SUBSET of logical people. "Some logical people are creative" only guarantees overlap SOMEWHERE in the logical-people group — but that overlap might fall entirely outside the engineers subset. Draw a Venn diagram mentally: Engineers ⊂ Logical, and Creative overlaps Logical somewhere — but nothing forces that overlap to touch Engineers specifically.

GATE FAVOURITE

GATE strategy
Always test syllogism conclusions by trying to draw a Venn diagram (or a counter-example scenario) where the premises hold true but the conclusion is FALSE. If you can construct even one such scenario, the conclusion is invalid — no matter how "intuitively likely" it feels.

Analogy

NOTES

Find the relationship between the first pair, then apply the SAME relationship to find the missing term.

WORKED EXAMPLE

Worked numerical

Q: Doctor : Hospital :: Teacher : ?
A: School. The relationship is "person : primary workplace". A doctor's primary workplace is a hospital; a teacher's primary workplace is a school.

WORKED EXAMPLE

Worked numerical

Q: 3 : 9 :: 5 : ? :: 7 : 49
A: 25. The relationship is "n : n²" (3²=9, 5²=25, 7²=49) — always verify your rule against BOTH given pairs before applying it to the unknown one.

Numerical relations & reasoning

WORKED EXAMPLE

Worked numerical — coding

Q: If CODE is written as DPEF (each letter shifted +1), how is WORD written?
A: Shift each letter forward by 1: W→X, O→P, R→S, D→E → **XPSE**.

WORKED EXAMPLE

Worked numerical — blood relations

Q: "A is B's father. B is C's sister. D is C's mother." How is A related to D?
A: B and C are siblings (B is C's sister, so B is female, C's sibling). A is their father, D is their mother. So **A is D's husband** (both are parents of the same children, B and C).

Seating arrangements — method

KEY POINT

Steps: (1) draw the skeleton — a row or a circle, and note which way people **face**; (2) fix the most constrained person first; (3) place all definite clues before any conditional ones; (4) branch only when forced; (5) re-check every clue against the final arrangement.

Counting: n people in a row → n! arrangements; around a circle → (n − 1)! (rotations are the same).

GATE FAVOURITE

GATE trap

In a circle, people facing the **centre** have their *left* neighbour clockwise from them; if they face **outward**, left and right swap. A row facing north has left = west and right = east. Always read which way the people face before using "left" or "right".

WORKED EXAMPLE

Worked example — row

Q: P, Q, R and S sit in a row facing north. S is at the extreme right. Q sits immediately to the left of S. R is not at either end. Who is at the extreme left, and what is the full order (left to right)?
A: S is in seat 4 and Q in seat 3. R is not at an end, so R is in seat 2 (seat 3 is taken). P takes seat 1. Order: **P, R, Q, S**; P is at the extreme left.

Rankings & ordering

KEY POINT

```
Total in a row or class = (rank from one end) + (rank from the other end) − 1
People between two positions = |difference of ranks| − 1
If two people swap places, only the ranks of those two people change.
```

WORKED EXAMPLE

Worked numerical — rank from both ends

Q: Meera is 12th from the top and 25th from the bottom of her class. How many students are there?
A: 12 + 25 − 1 = **36**.

WORKED EXAMPLE

Worked numerical — exchange of positions

Q: In a row, Arun is 7th from the left and Bala is 12th from the right. After they exchange seats, Arun is 22nd from the left. How many are in the row?
A: Arun now sits in Bala's old seat, so Bala's old seat is 22nd from the left and 12th from the right. Total = 22 + 12 − 1 = **33**.

Truth-teller / liar puzzles

MUST REMEMBER

Method: each person is a truth-teller (T) or a liar (L). Assume one person's type, follow what their statement forces on the others, and discard the case at the first contradiction. With 3 people there are only $2^3 = 8$ cases, so a small table is safe. Remember: a liar's statement must be **false as a whole** — the negation of "A and B are both liars" is "at least one of A, B is truthful".

WORKED EXAMPLE

Worked example — exactly one liar

Q: One of P, Q, R, S stole a watch. P: "I did not steal it." Q: "R stole it." R: "S stole it." S: "I did not steal it." Exactly one of the four is lying. Who stole it?
A: Test each suspect and count the false statements:

```
thief P → P, Q, R lie  (3 liars)     thief Q → Q, R lie  (2 liars)
thief R → only R lies  (1 liar) ✔    thief S → Q, S lie  (2 liars)
```

Only "R stole it" gives exactly one liar, so **R** is the thief.

Direction sense & instruction following

KEY POINT

Track position on a grid: north = +y, east = +x. A **right** turn moves the facing clockwise (N → E → S → W), a **left** turn anticlockwise. Distance from the start = $\sqrt{(\Delta x^2 + \Delta y^2)}$; direction = the direction of the displacement vector. At sunrise your shadow points west.

WORKED EXAMPLE

Worked numerical — walk with turns

Q: A walks 5 km north, turns right and walks 12 km, turns right and walks 5 km, then turns left and walks 3 km. How far, and in which direction, is A from the start?
A:

```
start (0,0)  → 5 km N  → (0,5)
turn right (facing E) → 12 km → (12,5)
turn right (facing S) →  5 km → (12,0)
turn left  (facing E) →  3 km → (15,0)
```

A is **15 km due east** of the start.

6 Spatial Aptitude

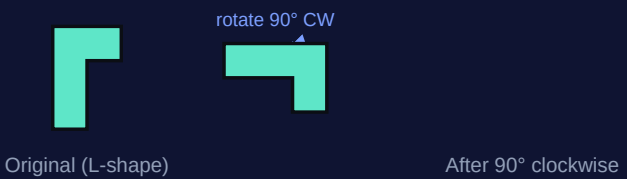
The five core transformations

REFERENCE TABLE

Transformation	What changes
Translation	Shape slides to a new position — size, orientation, and shape stay identical
Rotation	Shape turns around a fixed point by some angle (90°, 180°, etc.) — size stays the same, orientation changes
Scaling	Shape grows/shrinks uniformly — orientation stays the same, size changes
Mirroring (reflection)	Shape is flipped across a line — becomes a mirror image (left-right or up-down reversed)
Assembling/ Grouping	Multiple pieces combine to form a larger recognizable shape (jigsaw-style)

Rotation example

DIAGRAM



NOTES

Notice the L-shape's "foot" that pointed DOWN-RIGHT now points DOWN-LEFT after a 90° clockwise turn — track which specific corner/edge moves where, rather than trying to visualize the whole shape at once.

Mirror images

COMMON TRAP

Key rule

A **mirror image** (reflected left-right, like a vertical mirror) flips horizontal orientation but keeps vertical orientation unchanged. A **water image** (reflected top-bottom, like a reflection in a pond) does the reverse — flips vertical but keeps horizontal unchanged. Letters like **A, H, I, M, O, T, U, V, W, X, Y** look identical in a vertical mirror (they're symmetric left-right); letters like **B, C, D, E, H, I, O, X** look identical in a water/horizontal reflection (symmetric top-bottom).

Paper folding & cutting

WORKED EXAMPLE

Worked numerical

Q: A square paper is folded in half twice (first vertically, then horizontally), and a small triangle is cut from the folded corner (the corner where all folds meet). How many holes appear when fully unfolded?
A: Each fold doubles the number of layers at that corner. 2 folds → 2×2 = **4 layers** at the folded corner → cutting through all layers at once produces **4 holes** when unfolded, arranged symmetrically (one in each quadrant of the original square).

Cube/net folding

NOTES

A flattened "net" of 6 squares folds into a cube. The key skill: identify which faces become OPPOSITE each other (opposite faces never share an edge in the net) versus ADJACENT. In a standard cross-shaped net, the four squares in the middle row all become adjacent side faces, while the top and bottom squares fold to become the cube's top and bottom — opposite each other.

Counting figures & embedded figures

KEY POINT

Count **by size**, never by eye: label the vertices, count all figures of the smallest size, then the next size up, and add. For an embedded (hidden) figure, rotate or reflect the target mentally and check it fits within the larger figure *without altering any line*.

REFERENCE TABLE

n × n grid	Squares = n(n+1)(2n+1)/6	Rectangles (all) = [n(n+1)/2]²
1	1	1
2	5	9
3	14	36
4	30	100
5	55	225

FORMULA / RULE

Triangles: if the base of a triangle is divided into n equal parts and every division point is joined to the opposite apex, the number of triangles is n(n+1)/2 (n = 5 gives 15). Non-square rectangles = all rectangles − squares.

WORKED EXAMPLE

Worked numerical — 4 × 4 grid

Q: How many squares and how many rectangles (squares included) are in a 4 × 4 grid?
A: Squares: 16 (1×1) + 9 (2×2) + 4 (3×3) + 1 (4×4) = **30**. Rectangles: choose 2 of the 5 vertical lines and 2 of the 5 horizontal lines → C(5,2) × C(5,2) = 10 × 10 = **100**, of which 70 are non-square.

Dice & cubes

KEY POINT

A cube has 6 faces, 8 corners and 12 edges, and **three faces meet at every corner**. So faces seen together in one view are always **adjacent — never opposite**. Ordinary dice have opposite faces summing to 7, but a GATE puzzle does not assume that unless it says so.

Method: (1) from each view list the neighbours of every face; (2) a face with four known neighbours has the one remaining face as its opposite; (3) remove numbers already paired and repeat; (4) use the clockwise order around a corner only if adjacency alone does not settle it.

WORKED EXAMPLE

Worked example — two views

Q: One view of a cube shows faces 1, 2 and 5 at a corner; another shows 1, 6 and 3 at a corner. Which face is opposite 1?
A: The views make 2, 5, 6 and 3 all neighbours of face 1. That is four of the five other faces, so the remaining face, **4**, is opposite 1.

Figure series, patterns & abstract reasoning

MUST REMEMBER

For a series of figures, **track one feature at a time**: (1) **rotation** per step (angle and direction), (2) **count** change (dots, lines, sides), (3) **shading / colour** toggling, (4) **position** moving around a cycle. Different features usually have **different periods**; the whole figure repeats after the LCM of the periods.

```
Rotation by θ per step repeats the direction when kθ is a
multiple of 360°  → k = 360/gcd(θ, 360)
Direction after k steps = (k × θ) mod 360 :  clockwise from
the start.  Up 0° · Right 90° · Down 180° · Left 270°
Odd-one-out figure: find the property that all but one share
(symmetry, number of sides, shaded area, closed/open).
```

WORKED EXAMPLE

Worked numerical

Q: An arrow starts pointing east and rotates 90° anticlockwise each step. Which way does it point after 7 steps?
A: 7 × 90 = 630; 630 mod 360 = 270° anticlockwise from east → east → north (90°) → west (180°) → **south** (270°).

7 Problem → Standard Method Playbook

METHOD

1. Percentage / ratio wording

Translate the statement into a base quantity → ratio/equation. For successive percentage changes, use multiplication of factors: `new = old × (1±r/100)`. For “more than / less than”, keep the denominator explicit.

METHOD

2. Remainders / divisibility

Reduce numbers modulo the divisor. For powers, look for cycles; for divisibility, factor the divisor and apply the relevant tests. Do not assume the remainder distributes over multiplication without taking modulo.

METHOD

3. Mixture / alligation

Set quantities around the mean concentration/price. For two components, `quantity of cheaper : quantity of dearer = (dearer−mean) : (mean−cheaper)`.

METHOD

4. Critical reasoning

Identify conclusion → premises → hidden assumption. For a necessary assumption, negate the option: if the argument collapses, it is necessary. For strengthen/weaken, ask whether the option changes the causal or logical link rather than merely repeating a premise.

METHOD

5. Verbal deduction / syllogism

Translate “all/no/some” into set inclusion/exclusion. Test whether the conclusion follows in *every* model satisfying the premises. “Some A are B” does not imply “some B are all A”.

METHOD

6. Seating / ranking

Draw slots. Fix absolute positions first, then relative constraints. For circular seating, remove rotational symmetry by fixing one person. For rankings, convert “from left/right” into a common coordinate before combining.

METHOD

7. Spatial / cube / dice

Track adjacency and opposites. Faces seen together are adjacent. For cube rotations, preserve opposite pairs; for counting figures, count by size systematically rather than visually guessing.

METHOD

8. Speed / time / work

Convert to one unit system (km/h ↔ m/s: ×5/18). For trains add the lengths and use relative speed; for work questions take total work = LCM of the days and compute units per day. Equal-distance round trips use 2ab/(a+b).

METHOD

9. Sets, DI and estimation

Venn: fill from the triple intersection outward. DI: write the formula for the quantity first (profit % is on cost) and separate percent from percentage points. Estimation: round, check the order of magnitude, then eliminate options.

METHOD

10. Figure series

Treat each feature (rotation, count, shading, position) as its own sequence and find its period; the combined period is the LCM.

8 🎯 Representative GATE-Style Worked Problems

NOTE

Number system

Q: Find the remainder when `7^100` is divided by 13.
A: Powers of 7 modulo 13 cycle: `7, 10, 5, 9, 11, 12, 6, 3, 8, 4, 2, 1`, period 12. Since 100 ≡ 4 (mod 12), remainder = **9**.

NOTE

Mixture

Q: Rice costing ₹40/kg and ₹60/kg is mixed to obtain a mixture costing ₹45/kg. Find the ratio of quantities.
A: By alligation, cheaper : dearer = (60−45):(45−40) = 15:5 = **3:1**.

NOTE

Critical reasoning

Q: “The new study method improved scores because students who used it scored higher.” Which observation most weakens the conclusion?
A: If students who chose the method were already higher-performing before using it, the observed difference has an alternative explanation. That directly weakens the causal claim.

NOTE

Ranking

Q: A is 12th from the top and 19th from the bottom. How many people are there?
A: Total = 12 + 19 − 1 = **30**. Subtract 1 because A is counted from both ends.

NOTE

Calendar

Q: If 1 January is Monday in a non-leap year, what day is 1 March?
A: January + February = 31 + 28 = 59 days. 59 mod 7 = 3, so move three weekdays from Monday → **Thursday**.

NOTE

Spatial counting

Q: How many squares are in a 4×4 grid?
A: Sum `1²+2²+3²+4² = 30`. Equivalently `n(n+1)(2n+1)/6` for n=4 → **30**.

RULES & FACTS

1. 60 km/h; 13.5 s
2. 54 km/h; 12 s
3. 54 km/h; 13.5 s
4. 54 km/h; 10.8 s

FORMULA / WORKING

```
Distance while crossing the platform = 180 + 270 = 450 m in 30 s → speed = 15 m/s = 15 × 18/5 = 54 km/h
Man walking the same way: relative speed = 54 − 6 = 48 km/h = 48 × 5/18 = 40/3 m/s
Time to pass him = train length / relative speed = 180 ÷ (40/3) = 13.5 s
(12 s ignores the man's motion; 10.8 s wrongly adds the speeds, as if he were walking towards the train.)
```

NOTES

Correct: Option 3 — **54 km/h; 13.5 s**.

RULES & FACTS

1. 9 km/h; 9 h
2. 12 km/h; 8 h 20 min
3. 10 km/h; 7 h 30 min
4. 10 km/h; 8 h 20 min

FORMULA / WORKING

```
Downstream d = 24/2 = 12 km/h;  upstream u = 24/3 = 8 km/h
Still water b = (d + u)/2 = 10 km/h;  stream s = (d − u)/2 = 2 km/h
Time = 40/12 + 40/8 = 3 h 20 min + 5 h = 8 h 20 min
```

NOTES

Correct: Option 4 — **10 km/h** and **8 h 20 min**.

RULES & FACTS

1. 9 days; ₹1,500
2. 9 days; ₹1,800
3. 8 days; ₹1,500
4. 10 days; ₹1,500

FORMULA / WORKING

```
Let the job take T days. B works all T days; A works (T − 3) days.
Take the job = 36 units → A does 3 units/day, B does 2 units/day.
3(T − 3) + 2T = 36  →  5T = 45  →  T = 9 days
A's work = 3 × 6 = 18 units, B's work = 2 × 9 = 18 units → equal shares of the wages: A gets ₹1,500.
```

NOTES

Correct: Option 1 — **9 days**; A's share **₹1,500**.

RULES & FACTS

1. 42 years
2. 48 years
3. 36 years
4. 30 years

FORMULA / WORKING

```
Son = x, father = 3x.  After 12 years:  3x + 12 = 2(x + 12)  →  3x + 12 = 2x + 24  →  x = 12
Father = 3 × 12 = 36.   Check: in 12 years 48 and 24 → 48 = 2 × 24 ✓
```

NOTES

Correct: Option 3 — **36 years**.

RULES & FACTS

1. 60 exactly one; 0 none
2. 70 exactly one; 10 none
3. 60 exactly one; 10 none
4. 50 exactly one; 20 none

FORMULA / WORKING

```
Only M∩P = 25 − 10 = 15;  only P∩C = 20 − 10 = 10;  only M∩C = 15 − 10 = 5;  all three = 10
Only M = 60 − (15 + 5 + 10) = 30;  only P = 50 − (15 + 10 + 10) = 15;  only C = 40 − (5 + 10 + 10) = 15
Exactly one = 30 + 15 + 15 = 60
Union = 60 + 50 + 40 − 25 − 20 − 15 + 10 = 100  →  none = 110 − 100 = 10
```

NOTES

Correct: Option 3 — **60** study exactly one subject; **10** study none.

RULES & FACTS

1. 20 %; ₹100
2. 13.4 %; ₹67
3. 14 %; ₹70
4. 12 %; ₹60

FORMULA / WORKING

```
Marked price = 500 × 1.40 = 700
After the discounts: SP = 700 × 0.9 × 0.9 = 567   (one 19 % discount, not 20 %)
Profit = 567 − 500 = ₹67  →  67/500 = 13.4 %
```

NOTES

Correct: Option 2 — **13.4 %**, profit **₹67**.

RULES & FACTS

1. 2023; 40 %
2. 2019; 66.7 %
3. 2019; 40 %
4. 2023; 66.7 %

FORMULA / WORKING

```
Profit %: 2019 = 15/30 = 50 %;  2020 = 8/42 ≈ 19.0 %;  2021 = 5/40 = 12.5 %;  2022 = 12/48 = 25 %;  2023 = 20/55 ≈ 36.4 %
Highest percentage: 2019 (although 2023 has the largest absolute profit, ₹20 crore)
Revenue growth 2021 → 2023 = (75 − 45)/45 = 30/45 = 66.7 %   (the base is the earlier year, 2021)
```

NOTES

Correct: Option 2 — **2019** and **66.7 %**.

RULES & FACTS

1. Down; step 8
2. Left; step 4
3. Left; step 8
4. Right; step 8

FORMULA / WORKING

```
After 10 steps: 10 × 135 = 1350°;  1350 mod 360 = 270° clockwise from up → up 0°, right 90°, down 180°, left 270° → LEFT
First return to up: 135k must be a multiple of 360 → k = 360/gcd(135, 360) = 360/45 = 8   (8 × 135 = 1080 = 3 × 360)
```

NOTES

Correct: Option 3 — **Left**, and the arrow first points up again at **step 8**.

9 Last-minute one-liners

LAST-MINUTE · FORMULAS & RULES

- Ranking total = rank from top + rank from bottom − 1; the same person is counted from both ends.
- Alligation ratio = (dearer−mean):(mean−cheaper).
- Squares in an n×n grid = n(n+1)(2n+1)/6; rectangles = [n(n+1)/2]².
- Percentage change = (new-old)/old × 100%; successive percentage changes are multiplicative, not simply additive.
- **For speed, distance, and time:** distance = speed × time; use consistent units before calculation.
- Simple interest = PRT/100; compound interest depends on compounding frequency and period.

LAST-MINUTE · MUST REMEMBER

- **Trains:** pole = L/v; platform = (L+P)/v; two trains = (L_1+L_2)/relative speed (difference if same direction, sum if opposite).
- **Venn (3 sets):** fill from the triple overlap outward; $|A \cup B \cup C|$ = Σ singles − Σ pairs + triple.
- In aptitude, translate the wording into a mathematical model before calculating; avoid pattern-matching without checking constraints.
- For ratio questions, multiply all ratio parts by the same positive factor; ratios do not by themselves give absolute quantities.
- In probability, define the sample space carefully; mutually exclusive and independent are different concepts.
- For data interpretation, estimate first when options are widely separated; then calculate only as precisely as necessary.
- Geometry questions should begin by identifying the relevant invariant/standard formula rather than expanding algebra blindly.
- **Critical reasoning:** a weakener offers an alternative cause, a strengthener rules one out, and a necessary assumption collapses the argument when negated.

LAST-MINUTE · COMMON TRAPS

- In seating/cube questions, preserve relative constraints; do not infer adjacency merely from visual placement.
- Average of averages is not generally the average of the group averages unless group sizes are equal or weighted appropriately.
- $P(A \cup B)=P(A)+P(B)-P(A \cap B)$; do not omit the intersection unless events are mutually exclusive.
- 'All A are B' does not imply 'All B are A'; only the contrapositive (not B → not A) is guaranteed.

LAST-MINUTE · SHORTCUTS

- **Estimation:** round to friendly numbers, check the order of magnitude, eliminate options. Unit digits cycle (7: 7,9,3,1).
- Use approximation, ratio comparison, cancellation, and option elimination whenever the question permits it.

LAST-MINUTE · GATE FAVOURITES

- **Figure series:** track rotation, count, shading and position separately; combined period = LCM of the periods.
- **Number-system questions:** reduce powers modulo the divisor and look for a cycle before calculating large exponents.